

## Team Information

| Name    | Role / Contribution                     | Files   | Hours |
|---------|-----------------------------------------|---------|-------|
| Will W. | The log and peer table logic and output | node.py | 4     |
| Liam O. | Message handling and packet parsing     | node.py | 5     |
| Nick A. | Broadcasting send/receive               | node.py | 5     |

## 1. System Setup

- Hardware: Raspberry Pi 3
- Network: Wi-Fi 10.129.255.255
- UDP Port: 5000

## 2. Protocol Summary

- Broadcast: 5 s
- Ping: 15 s
- Timeout: 3 s
- Removal: 30 s

## 3. Results

| Metric         | Value   | Notes                 |
|----------------|---------|-----------------------|
| Avg RTT        | 59.5 ms | from PING/PONG        |
| Convergence    | <1sec   | Form ping/pong        |
| Discovery time | <1sec   | until full table sync |

## 4. Observations

- Nodes discovered: 2
- Average discovery time: <1 seconds
- Average RTT between peers: 59.5 ms
- Packet loss rate (approx.): 0 %
- Peer tables converged after: <1 seconds

## 4. Logs

```
2025-11-03 14:24:57,863 [INFO] | INIT Node pi1 at 10.129.133.30:5000
2025-11-03 14:24:57,863 [INFO] | [TABLE] 0 active peers
2025-11-03 14:24:57,863 [INFO] | START Threads: listener, broadcaster, heartbeat, summary
2025-11-03 14:24:58,104 [INFO] | [SYNC] Added pi2 (10.129.207.40)
2025-11-03 14:24:58,310 [INFO] | [SYNC] Added pi3 (10.129.202.9)
2025-11-03 14:25:07,869 [INFO] | [TABLE] 2 active peers
2025-11-03 14:25:12,994 [INFO] | [PING] RTT=14ms to pi2
2025-11-03 14:25:13,070 [INFO] | [PING] RTT=90ms to pi3
2025-11-03 14:25:17,874 [INFO] | [TABLE] 2 active peers
2025-11-03 14:25:27,880 [INFO] | [TABLE] 2 active peers
2025-11-03 14:25:28,133 [INFO] | [PING] RTT=7ms to pi2
2025-11-03 14:25:28,253 [INFO] | [PING] RTT=127ms to pi3
2025-11-03 14:25:29,116 [INFO] | STOP Node stopped cleanly
```

## 5. Peer Tables

```
peers = {
  "pi2":{
    "ip":"10.129.207.40",
    "port":5000,
    "last_seen":1762201528,
    "status":"active"
  },
  "pi3":{
    "ip":"10.129.202.9",
    "port":5000,
    "last_seen":1762201528,
    "status":"active"
  }
}
```

```
peers_snapshot_pi3.txt
peers = {
  "pi1":{
    "ip":"10.129.133.30",
    "port":5000,
    "last_seen":1762201531,
    "status":"active"
  },
  "pi2":{
    "ip":"10.129.207.40",
    "port":5000,
    "last_seen":1762201532,
    "status":"active"
  }
}
```

```
logs > peers_snapshot_pi2.txt
1  peers = {
2    "pi3":{
3      "ip":"10.129.202.9",
4      "port":5000,
5      "last_seen":1762201529,
6      "status":"active"
7    },
8    "pi1":{
9      "ip":"10.129.133.30",
10     "port":5000,
11     "last_seen":1762201528,
12     "status":"active"
13   }
14 }
15
```

## 6. Discussion

- It seems like everything we needed to implement for this section works as intended.

## 7. Next Steps

- Further testing to ensure all parts of the program work including removal from table. Begin distributed learning for next deliverable